

Junghwan Yim

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RESEARCH INTERESTS

Agentic digital twins; robotics security; sensing and applied machine learning. Developing methods that connect physical observations, virtual models, and robot behavior, with an emphasis on implementation and evaluation.

EDUCATION

Georgia Institute of Technology Ph.D. in Mechanical Engineering, in progress	Jul 2025 - Present
University at Buffalo M.S., Physics; B.S., Computational Physics; B.S., Computer Science	2016 - 2022
State University of New York at Fredonia Bachelor's degree, Physics with Pre-Law Bachelor's degree, Business Administration with Music Industry	2013 - 2016

PROFESSIONAL EXPERIENCE

Hyundai Motor Group - Meta Assembly AI Solution Architect (Lead), Industry Collaboration / Client Project Architected a Transformer world model on an NVIDIA Isaac Sim factory digital twin for AMR routing and logistics. Built a departmental multi-agent LLM platform on Kubernetes with MCP data access. Developed a Neo4j knowledge ontology and real-time consistency validation, reducing the daily production-issue rate by 80% . Owned data pipelines, deployment, GPU scheduling, and observability for 24/7 operations .	2025 - Present
SK C&C USA, Inc. Senior Solution Engineer - AI Robotics Specialist Designed AI, robotics, and digital twin solutions from customer requirements through PoC and production handoff, generating approximately \$35M in potential sales pipeline . Led PROMAT 2025 and digital marketing initiatives with a \$30K budget , securing 300+ leads . Architected AMR/AGV, RFID, and inspection integrations with MES/WMS/ERP systems; delivered customer workshops and onboarding.	Sep 2024 - Aug 2025
Automation Project Engineer Engineered automation for the Ford-SK On battery factory across software, PLCs, vision, and AMR/AGV control. Built a RAG-based engineering-knowledge platform delivering approximately \$20M in cost savings through reduced rework and faster onboarding. Developed AI inspection, drawing generation, 3D installation verification, and logistics simulations.	Mar 2023 - Aug 2024
Hyundai AutoEver America Software Test Engineer Developed AI-assisted navigation testing for the Genesis lineup using sign and vehicle-state recognition. Contributed to HD maps supporting autonomous-driving and ADAS functions.	Oct 2022 - Mar 2023
Republic of Korea Army - Cyber Operation Command Software Engineer / Data Scientist Architected and operated a petabyte-scale distributed intelligence platform for data collection, NLP classification, and entity resolution using Hadoop, Spark, Kafka, and ELK. Built React/Django interfaces and Celery/RabbitMQ/Redis pipelines. Received 2 Meritorious Service Medals for software development and the K-Startup Contest (2019-2020).	May 2019 - Dec 2020

RESEARCH EXPERIENCE

Georgia Institute of Technology Graduate Research Assistant Built multi-node, multi-GPU training pipelines and multiscale simulation frameworks combining learned interatomic potentials with DFT. Developed synchronized LiDAR/camera/radar testbeds. Co-authored SeeTraceAct with Prof. Zsolt Kira's group, fine-tuning VLA action policies on real Franka Panda hardware.	Aug 2025 - Present
University at Buffalo - Biometrics Lab Graduate Student Researcher Developed authentication models using phone and watch IMU signals, transfer functions, CycleGAN, signal processing, and machine learning. Built phone and watch applications for data collection, Bluetooth-connected sensor monitoring, and model-based authentication.	Jun 2022 - Jun 2024
University at Buffalo - CAVAS Lab Graduate Student Researcher Led a traffic-sign object detection team and built a traffic-sign knowledge graph. Trained YOLO and RCNN models on Mapillary traffic-sign data, with the best model achieving 70% accuracy . Implemented A2C and DQN agents in synchronous and asynchronous CARLA simulation, and collaborated on a ROS package mapping detected signs and distances into an HD map.	Jan 2021 - Jun 2024

University at Buffalo - Condensed Matter Physics Lab

Jun 2016 - Jun 2022

Graduate Student Researcher

Modeled electron scattering and graphene transport using the Caldeira-Leggett model and semiclassical Boltzmann transport equation. Implemented Python and C++ models with high-performance computing and MPI; visualized and evaluated models under varying conditions.

University at Buffalo - Blockchain Lab

May 2017 - May 2019

Graduate Student Researcher

Participated in developing a Solidity DApp for credit-information exchange in an M&T Bank hackathon case.

TEACHING

SUNY Fredonia

Aug 2015 - Jan 2016

Teaching Assistant, Physics Laboratories

Guided experiments and advised students on equipment and procedures across **4 physics lab courses**: University Physics I/II and College Physics I/II.

JOURNAL AND CONFERENCE PUBLICATIONS

- [1] Young-Dae Shim, **Junghwan Yim**, Nian Liu, Cheong Hoon Kwon, Hae-Jin Choi, Seung-Kyum Choi. Cross-Chemistry Meta-Learning with Frequency-Domain Augmentation for Robust Capacity Retention Prediction in Vanadium Redox Flow Batteries. *IEEE Access*, 2026. <https://doi.org/10.1109/ACCESS.2026.3737729>
- [2] Seokmin Choi, **Junghwan Yim**, Se Jun Kim, Yincheng Jin, Di Wu, Zhanpeng Jin. VibPath: Two-Factor Authentication with Your Hand's Vibration Response to Unlock Your Phone. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 2023. <https://doi.org/10.1145/3610894>
- [3] Seokmin Choi, **Junghwan Yim**, Yincheng Jin, Yang Gao, Jiyang Li, Zhanpeng Jin. EarPPG: Securing Your Identity with Your Ears. *ACM International Conference on Intelligent User Interfaces (IUI)*, 2023. <https://doi.org/10.1145/3581641.3584070>

WORKSHOP PAPER

- [4] Jaehyeon Son, Junhyun Kim, Kyle Kam, Jinhoo Kim, Seok Joon Kim, **Junghwan Yim**, Zsolt Kira. SeeTraceAct: Visibility-Aware Latent Planning from Cross-Embodiment Demonstration Videos. *SCALE Workshop @ ICML 2026*. <https://openreview.net/forum?id=xZbOwW72u5>

PREPRINT

- [5] Mainak Mallick, **Junghwan Yim**, Alankrit Gupta, Seung-Kyum Choi. Rad-R: A Raw-ADC Radar Dataset and Capture-Invariant SSM for Hardware-Fault Diagnosis. *arXiv preprint*, 2026. <https://arxiv.org/abs/2608.30896>

SELECTED ENGINEERING PROJECTS

Music NFTs and audience engagement

Feb 2022 · Chartmetric · Music analytics · Measure of Music 2022

A Measure of Music conference project examining music NFT releases using Chartmetric data. The analysis considered Shawn Mendes and Tory Lanez releases alongside follower and streaming trends.

Reinforcement learning agents in CARLA

Mar - May 2021 · CARLA · DQN · A2C

Autonomous driving agents using DQN and A2C in the CARLA simulator, with synchronous and asynchronous simulation environments.

Rumour detection with BERT and GCN

Feb - May 2021 · BERT · Graph convolutional networks · RumourEval 2019

A model for classifying discussion stances and detecting rumours in Twitter and Reddit data from RumourEval 2019. The project combines BERT with a three-layer graph convolutional network.

Achieved **67% stance-classification accuracy** and **70% rumour-detection accuracy** on RumourEval 2019. Selected as the best study in the Advanced Natural Language Processing class.

SELECTED HONORS AND AWARDS

Employee of the Year , SK C&C USA Recognition for AI-aided manufacturing and automation software engineering.	2023
Meritorious Service Medal , ROK Personnel Command Recognition in the K-Startup Contest.	May 2020
Meritorious Service Medal , ROK Cyber Operations Command Recognition for software development contributions.	Dec 2019
Dean's List , SUNY Fredonia	May/Dec 2014; May/Dec 2015
Alpha Lambda Delta , National Honor Society for First-Year Students · May 2014	
SUNY Global Scholarship , SUNY Fredonia · Dec 2013	

SELECTED CERTIFICATIONS AND TRAINING

RCR Basic Course - CITI Program · Aug 2025
AWS Partner: Sales Accreditation - Amazon Web Services · Jun 2025
Self-Driving Cars - University of Toronto · Oct 2023
Google Project Management - Google · Jun 2023
Data Intensive Computing - University at Buffalo · May 2019
Probabilistic Graphical Models - Stanford University · Oct 2019

TECHNICAL SKILLS AND LANGUAGES

Implementation: Python, C++, PyTorch, TensorFlow, ROS/ROS2, Isaac Sim, CARLA, React, Django, SQL.
Computing: MPI, NCCL, CUDA, distributed GPU training, Kubernetes, Hadoop, Kafka, Spark.
AI systems: World models, VLA, RAG, MCP, knowledge graphs, Neo4j, LangChain.
Languages: Korean (native/bilingual); English (professional working proficiency).